TRABAJO GRUPAL PIN2

Grupo 9

DevOps 2403

PIN2

Objetivos

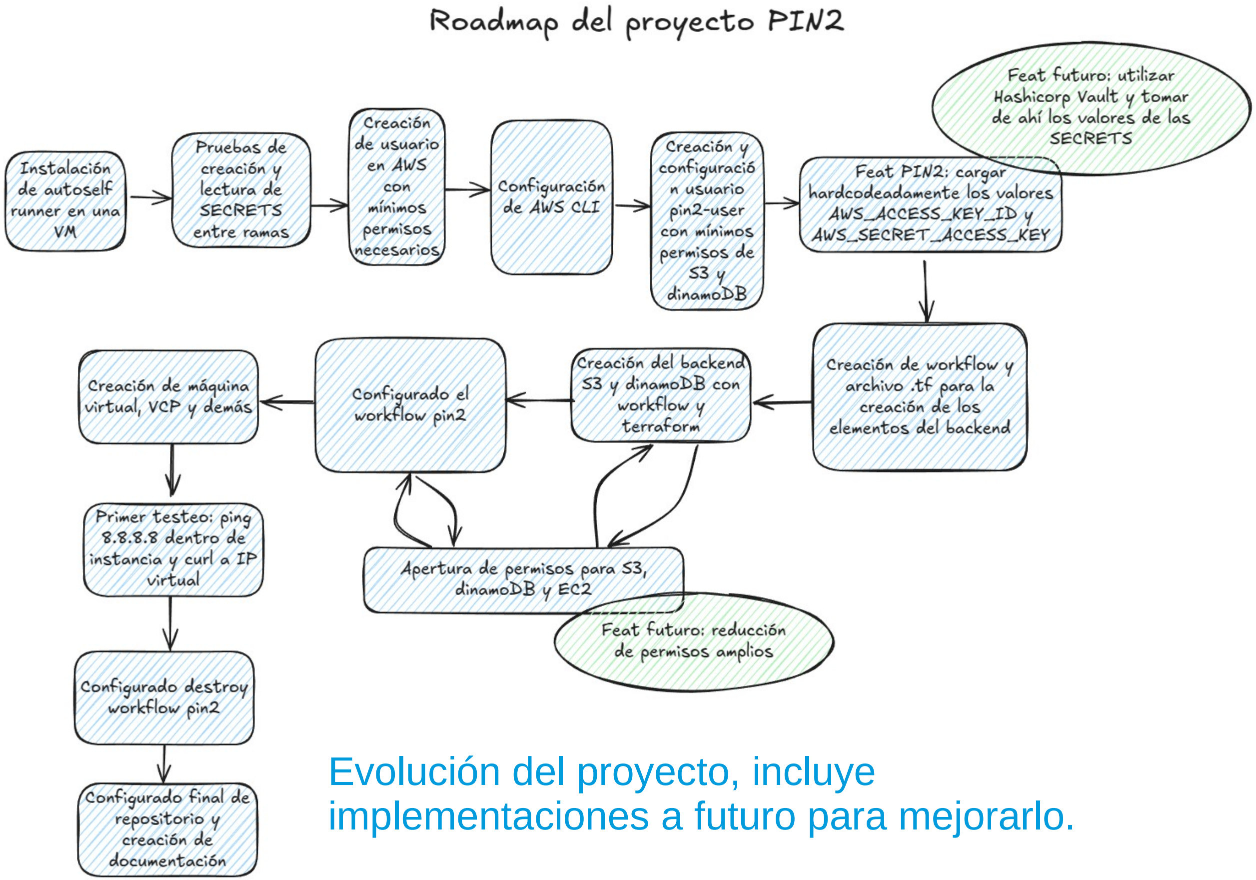
* Objetivo principal: a través de GitHub Workflow y terraform desplegar una VPC, IGW, tablas de rutas, subred, SG y una instancia EC2, que configure con un script un servicio de Apache, utilizando como backend un bucket S3 y una tabla dinamoDB.

* Objetivo secundario: a través de GitHub Workflow y terraform desplegar el backend.

Repositorio

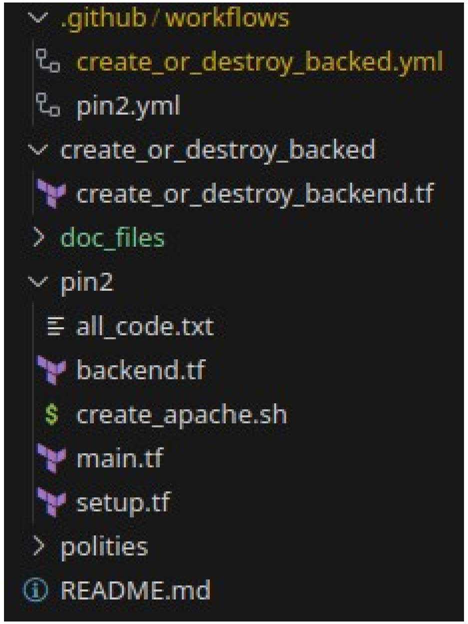
https://github.com/MauroPereira/PIN2_public

PIN2



Evolución del proyecto, incluye implementaciones a futuro para mejorarlo.

Existe dos GitHub workflows con sus archivos terraform separados por carpetas, “create_or_destroy_backend” para el objetivo secundario y “pin2” para el objetivo principal.



PIN2 – create or destroy backend

create_or_destroy_backend.tf ×

create_or_destroy_backed > create_or_destroy_backend.tf

```hcl
terraform {
  backend "local" {
    path = "terraform.tfstate"
  }
}

resource "aws_s3_bucket" "terraform_state_bucket" {
  bucket = "mundose222-mauro-a-pereira"
}

resource "aws_dynamodb_table" "terraform_lock_table" {
  name         = "terraformstatelock"
  billing_mode = "PAY_PER_REQUEST"
  hash_key     = "LockID"

  attribute {
    name = "LockID"
    type = "S"
  }

  timeouts {
    create = "5m"
  }
}
```

Para crear el backend se utiliza un tfstate local, para eliminar los recursos al tratarse de un workflow y al ser efímero, se reconstruye el archivo tfstate y con esa información se destruyen los recursos.

```yaml
- name: Importing Init
  id: importing
  run: |
    echo "Initializing Terraform..."
    terraform init

    echo "Importing existing S3 bucket..."
    terraform import aws_s3_bucket.terraform_state_bucket mundo

    echo "Importing existing DynamoDB table..."
    terraform import aws_dynamodb_table.terraform_lock_table te
  working-directory: create_or_destroy_backed
  env:
    AWS_ACCESS_KEY_ID: ${{ secrets.AWS_ACCESS_KEY_ID }}    Contex
    AWS_SECRET_ACCESS_KEY: ${{ secrets.AWS_SECRET_ACCESS_KEY }}
    AWS_DEFAULT_REGION: ${{ secrets.AWS_DEFAULT_REGION }}
  continue-on-error: false

- name: Display Terraform State
  id: display_tfstate
  run: |
    echo "Displaying generated terraform.tfstate file:"
    cat terraform.tfstate
  working-directory: create_or_destroy_backed
  env:
    AWS_ACCESS_KEY_ID: ${{ secrets.AWS_ACCESS_KEY_ID }}    Cont
    AWS_SECRET_ACCESS_KEY: ${{ secrets.AWS_SECRET_ACCESS_KEY }}
    AWS_DEFAULT_REGION: ${{ secrets.AWS_DEFAULT_REGION }}    Co
  continue-on-error: false
```

PIN2 – create or destroy backend - GH Actions

← Create or destroy S3 and dinamoDB backend

Create or destroy S3 and dinamoDB backend #10

Summary

Jobs

- terraform_create_backend
- terraform_destroy_backend

Run details

- Usage
- Workflow file

Manually triggered 2 days ago	Status	Total duration	Billable time	Artifacts
MauroPereira -o- f1c4183 test	Success	32s	1m	–

create_or_destroy_backed.yml

on: workflow_dispatch

- terraform_create_backend 23s
- terraform_destroy_backend 0s

PIN2 – create or destroy backend - GH Actions

Summary

Jobs

- terraform_create_backend
- terraform_destroy_backend

Run details

- Usage
- Workflow file

terraform_create_backend
succeeded 2 days ago in 23s

Search logs

Terraform Apply

```
      + logging (known after apply)

      + object_lock_configuration (known after apply)

      + replication_configuration (known after apply)

      + server_side_encryption_configuration (known after apply)

      + versioning (known after apply)

      + website (known after apply)
    }

Plan: 2 to add, 0 to change, 0 to destroy.
aws_dynamodb_table.terraform_lock_table: Creating...
aws_s3_bucket.terraform_state_bucket: Creating...
aws_s3_bucket.terraform_state_bucket: Creation complete after 1s [id=mundose222-mauro-a-pereira]
aws_dynamodb_table.terraform_lock_table: Creation complete after 7s [id=terraformstatelock]

Apply complete! Resources: 2 added, 0 changed, 0 destroyed.
::debug::Terraform exited with code 0.
```

Post Checkout

Complete job

PIN2 – pin2

Al haber creado el backed, este se utiliza para guardar la información de todos los recursos creados con pin2.

backend.tf ×

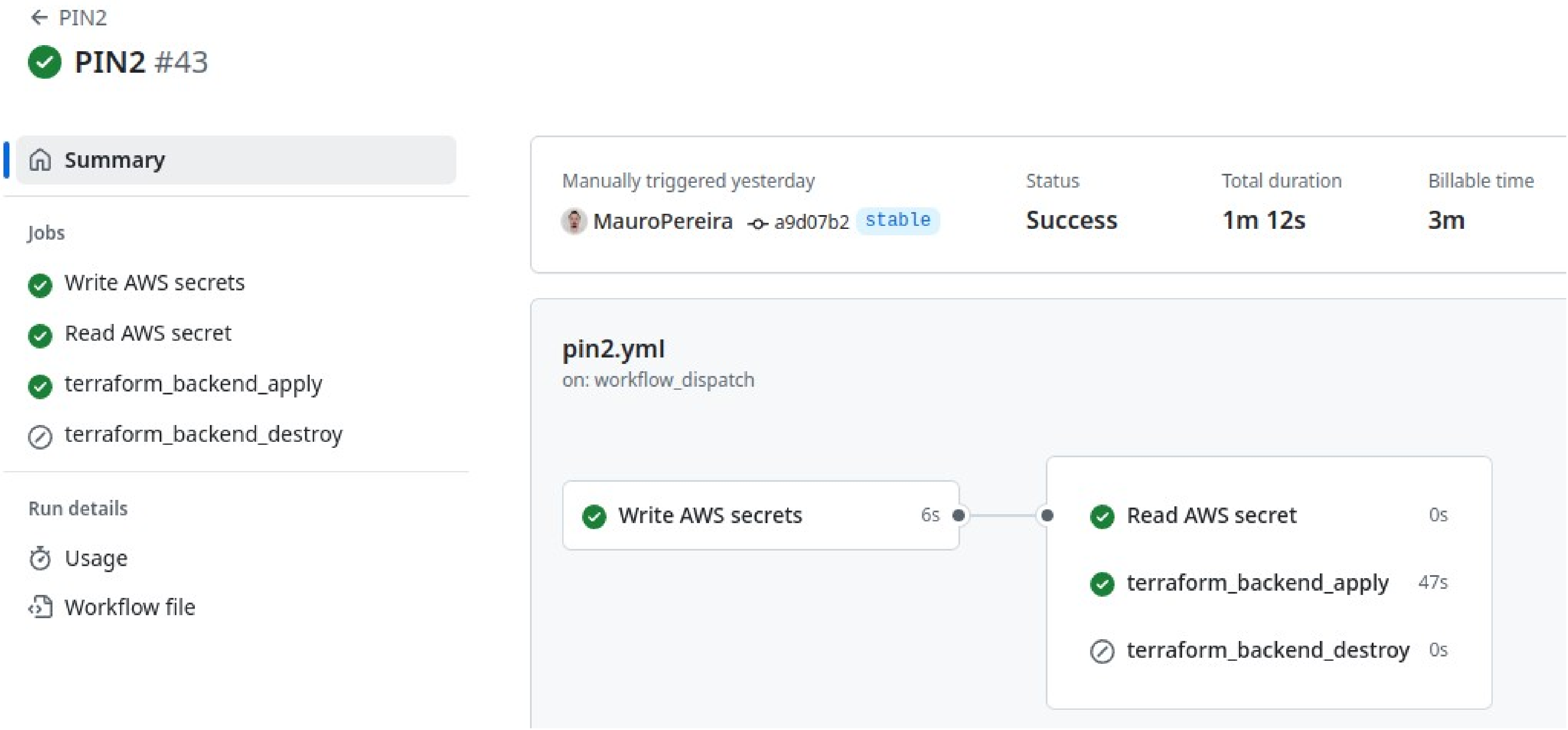
pin2 > backend.tf

```
terraform {
  backend "s3"{
    bucket                    = "mundose222-mauro-a-pereira"
    # region                    = "us-east-1"   # No necesario
    key                       = "backend.tfstate"
    dynamodb_table            = "terraformstatelock"
  }
}
```

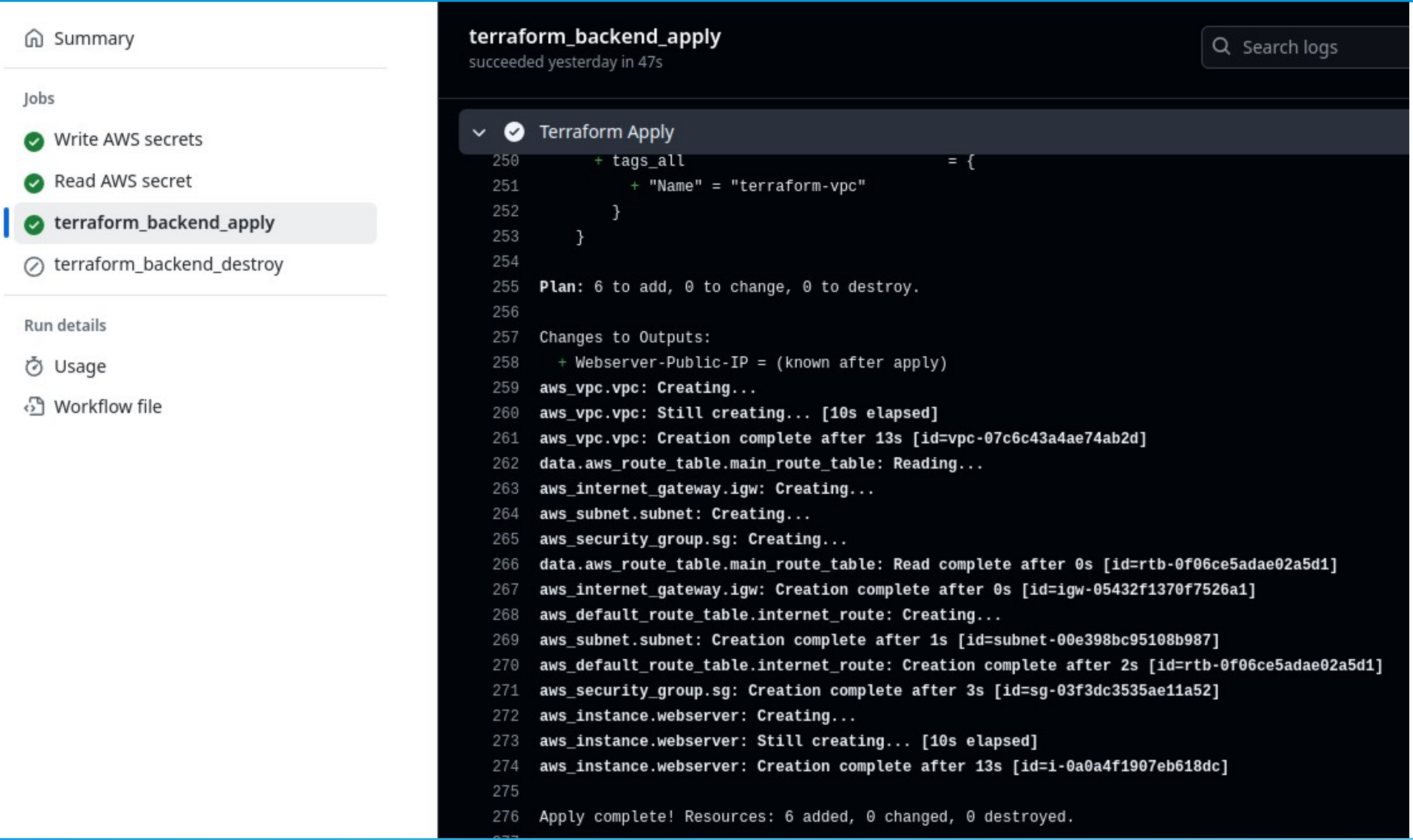
```
  workflow_dispatch:
    inputs:
      action:
        description: "Action to perform"
        type: choice
        options:
          - Create backend
          - Destroy backend
          - Write AWS secrets and Create backend
      aws_access_key_id:
        description: "AWS Access Key ID"
        required: true
      aws_secret_access_key:
        description: "AWS Secret Access Key"
        required: true
      aws_default_region:
        description: "AWS Default Region"
        default: "us-east-1"
        required: false

jobs:
  write_aws_secrets:
    if: ${{ (github.event_name == 'workflow_dispatch' && github.event.inputs.action == 'Write AWS secrets and Create backend')  }}
    name: "Write AWS secrets"
    runs-on: ubuntu-latest
    steps:
      - name: "Write AWS_ACCESS_KEY_ID"
        uses: hmanzur/actions-set-secret@v2.0.0
        with:
          name: "AWS_ACCESS_KEY_ID"
          value: ${{ github.event.inputs.aws_access_key_id }}
          token: ${{ secrets.REPO_SECRET_WRITE_TOKEN }}    Context access might be invalid: REPO_SECRET_WRITE_TOKEN

      - name: "Write AWS_SECRET_ACCESS_KEY"
        uses: hmanzur/actions-set-secret@v2.0.0
        with:
          name: "AWS_SECRET_ACCESS_KEY"
          value: ${{ github.event.inputs.aws_secret_access_key }}
          token: ${{ secrets.REPO_SECRET_WRITE_TOKEN }}    Context access might be invalid: REPO_SECRET_WRITE_TOKEN

      - name: "Write AWS_DEFAULT_REGION"
        uses: hmanzur/actions-set-secret@v2.0.0
        with:
          name: "AWS_DEFAULT_REGION"
          value: ${{ github.event.inputs.aws_default_region }}
          token: ${{ secrets.REPO_SECRET_WRITE_TOKEN }}    Context access might be invalid: REPO_SECRET_WRITE_TOKEN
```

Se ofrece una función adicional, se pueden crear los secrets necesarios para AWS en Actions.

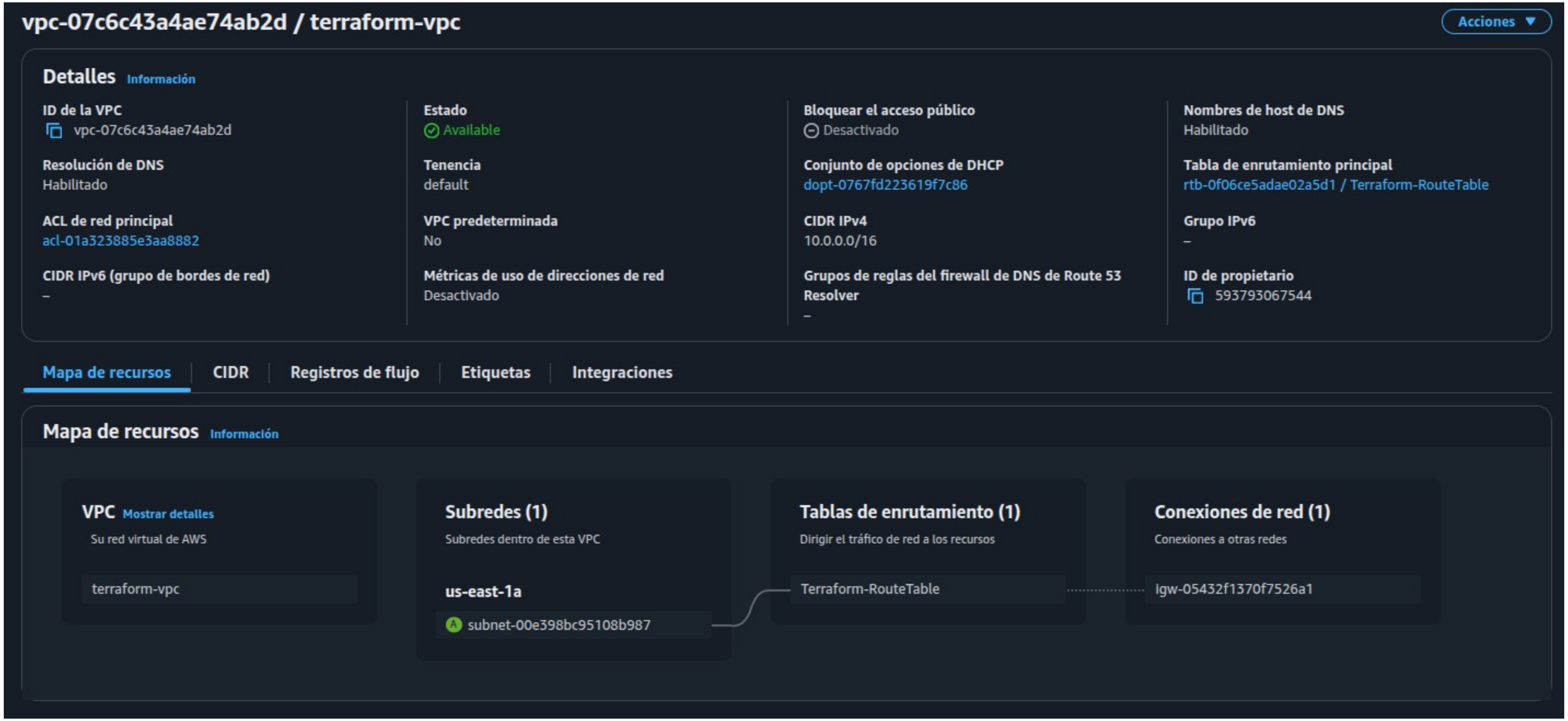
PIN2 – pin2 - GH Actions



PIN2 – pin2 - GH Actions

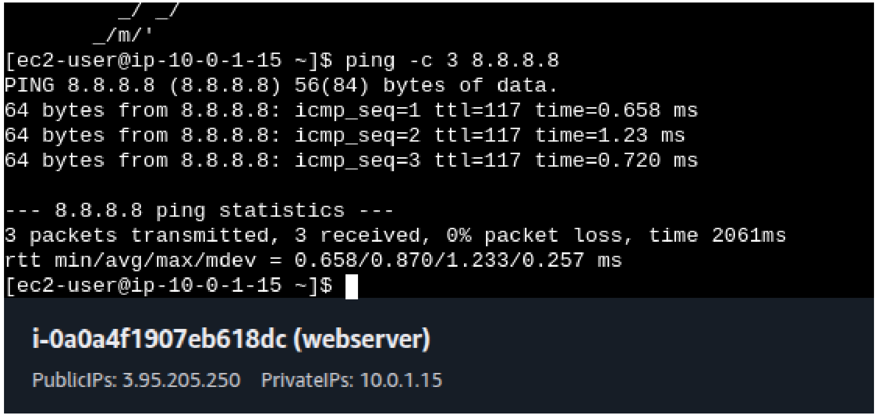


Chequeo final



Se pueden observar todos los recursos creados en la VPC.

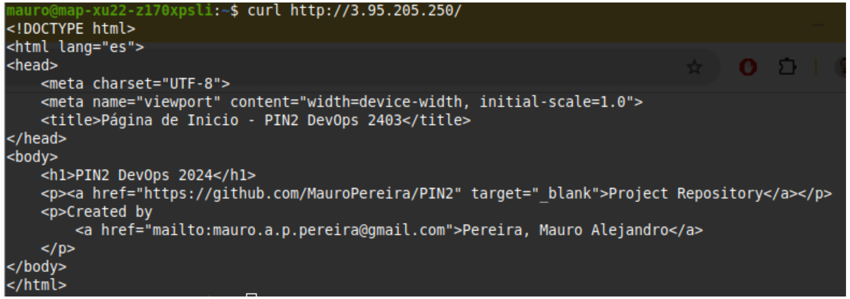
Chequeo final



```
        _/ _/
      _/m/'
[ec2-user@ip-10-0-1-15 ~]$ ping -c 3 8.8.8.8
PING 8.8.8.8 (8.8.8.8) 56(84) bytes of data.
64 bytes from 8.8.8.8: icmp_seq=1 ttl=117 time=0.658 ms
64 bytes from 8.8.8.8: icmp_seq=2 ttl=117 time=1.23 ms
64 bytes from 8.8.8.8: icmp_seq=3 ttl=117 time=0.720 ms

--- 8.8.8.8 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2061ms
rtt min/avg/max/mdev = 0.658/0.870/1.233/0.257 ms
[ec2-user@ip-10-0-1-15 ~]$

i-0a0a4f1907eb618dc (webserver)
PublicIPs: 3.95.205.250   PrivateIPs: 10.0.1.15
```

Desde adentro de la instancia EC2 efectivamente hay conexión a internet, lo que demuestra que el despliegue está correcto.

Desde una terminal cualquiera, fuera de la nube privada de AWS, se puede obtener la información de Apache.



```
mauro@map-xu22-z170xpsli:~$ curl http://3.95.205.250/
<!DOCTYPE html>
<html lang="es">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Página de Inicio - PIN2 DevOps 2403</title>
</head>
<body>
    <h1>PIN2 DevOps 2024</h1>
    <p><a href="https://github.com/MauroPereira/PIN2" target="_blank">Project Repository</a></p>
    <p>Created by
        <a href="mailto:mauro.a.p.pereira@gmail.com">Pereira, Mauro Alejandro</a>
    </p>
</body>
</html>
```